Coulomb’s law

antocciphysics | TEACHER KEY | CPA

Learning target: Relate electric force to charge magnitude and separation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 17; 3 course days later.*
The farther target takes 3 times as long because t=d/v.

02. Explain a worked example (worked)
Equal opposite repulsive forces; larger separation yields weaker force by factor 4.

03. Look and explain (model)
Attraction remains; each force becomes one fourth.

04. Build the solution (practice)
3 N.

05. Use the idea (reason)
No; the interaction forces have equal magnitudes by Newton’s third law.

06. Choose a method (mixed) *Review: Unit 6, Day 8; 12 course days later.*
Parallel oscillation; a longitudinal pressure wave.

07. Exit check (exit)
It triples.

08. STRETCH: Apply it another way (stretch)
12×2/4=6 N; sign relationship unchanged unless a charge sign changes.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 17; 3 course days later.*
v≈17.32 m/s; t≈288.7 s. Requires the long-wave shallow-water regime and simplified depth; not a damage forecast.

02. Explain a worked example (worked)
Equal opposite repulsive forces; larger separation yields weaker force by factor 4.

03. Represent and interpret (model)
Attraction remains; each force becomes one fourth.

04. Solve and justify (practice)
0.60 N; attraction.

05. Analyze the reasoning (reason)
No; the interaction forces have equal magnitudes by Newton’s third law.

06. Choose a method (mixed) *Review: Unit 6, Day 8; 12 course days later.*
240 cycles; T≈0.001667 s.

07. Exit check (exit)
4/9 of the original force.

08. STRETCH: Apply it another way (stretch)
0.6 N attractive; at 0.9 m, force 0.6/9≈0.0667 N.